

The Curvature Tensor

Let none ignorant of geometry enter my door.

—INSCRIPTION OVER THE ENTRANCE TO PLATO'S ACADEMY

The Riemann curvature tensor, or simply the curvature tensor, is a cornerstone of Riemann's theory of non-Euclidean geometry. A mixed tensor of rank 4, it serves as a measure of the departures of any space from simple Euclidean, or pseudo-Euclidean form. If any single component of the curvature tensor is nonvanishing, the space is distorted, or, more informally, "curved."¹ As befits its tensorial status, this statement is coordinate independent. The curvature tensor may be contracted on an upper and lower index to form a tensor of rank 2, known as the Ricci tensor. The Ricci tensor is linear in the second derivatives of $g_{\mu\nu}$ and is a key component of the Einstein field equations for gravity. In this chapter, we will see how the curvature tensor emerges naturally in the formulation of Riemannian geometry and how the Ricci tensor is incorporated in the final form of the Einstein field equations.

5.1 Commutation rule for covariant derivatives

The covariant derivative shares many properties with the ordinary partial derivative: it is a linear operator, it obeys the Leibniz rule, and it allows true tensor status to be bestowed upon "upgraded" partial derivatives (or upon ordinary derivatives with respect to a scalar parameter) in any coordinates. A natural question arises. Normally, partial derivatives operating on a function commute: the order in which they are taken does not matter, provided that appropriate smoothness conditions are present. Does the same commutation law work for covariant derivatives? Does $(V^\mu_{;\sigma})_{;\tau}$ equal $(V^\mu_{;\tau})_{;\sigma}$?

Just do it.

$$V^\mu_{;\sigma} = \frac{\partial V^\mu}{\partial x^\sigma} + \Gamma^\mu_{\nu\sigma} V^\nu \equiv T^\mu_{\sigma}. \quad (5.1)$$

Then,

$$T^\mu_{\sigma;\tau} = \frac{\partial T^\mu_{\sigma}}{\partial x^\tau} + \Gamma^\mu_{\nu\tau} T^\nu_{\sigma} - \Gamma^\nu_{\sigma\tau} T^\mu_{\nu}, \quad (5.2)$$

¹*Curvature* is a somewhat misleading mathematical label, like *imaginary* numbers. The name implies an external dimension into which the space is curved or embedded, an unnecessary complication.

or

$$T^\mu_{\sigma;\tau} = \frac{\partial^2 V^\mu}{\partial x^\tau \partial x^\sigma} + \frac{\partial}{\partial x^\tau} \left(\Gamma^\mu_{\lambda\sigma} V^\lambda \right) + \Gamma^\mu_{\nu\tau} \left(\frac{\partial V^\nu}{\partial x^\sigma} + \Gamma^\nu_{\lambda\sigma} V^\lambda \right) - \Gamma^\nu_{\sigma\tau} \left(\frac{\partial V^\mu}{\partial x^\nu} + \Gamma^\mu_{\lambda\nu} V^\lambda \right). \quad (5.3)$$

The first term and the last group (proportional to $\Gamma^\nu_{\sigma\tau}$) are manifestly symmetric in σ and τ , and so will vanish when the same calculation is done with these indices reversed and then subtracted. A bit of inspection shows that the same is true for the two remaining terms that are proportional to a partial derivative of V^μ . The residual terms from taking the covariant derivative commutator are then

$$T^\mu_{\sigma;\tau} - T^\mu_{\tau;\sigma} = \left[\frac{\partial \Gamma^\mu_{\lambda\sigma}}{\partial x^\tau} - \frac{\partial \Gamma^\mu_{\lambda\tau}}{\partial x^\sigma} + \Gamma^\mu_{\nu\tau} \Gamma^\nu_{\lambda\sigma} - \Gamma^\mu_{\nu\sigma} \Gamma^\nu_{\lambda\tau} \right] V^\lambda, \quad (5.4)$$

which we may write as

$$(V^\mu_{;\sigma})_{;\tau} - (V^\mu_{;\tau})_{;\sigma} = T^\mu_{\sigma;\tau} - T^\mu_{\tau;\sigma} \equiv R^\mu_{\lambda\sigma\tau} V^\lambda. \quad (5.5)$$

Now, both sides of this equation must be tensors, and V^λ is itself an arbitrary vector. This means that the quantity $R^\mu_{\lambda\sigma\tau}$ must transform its coordinate indices as a tensor. This statement may be explicitly verified by a lengthy calculation, but the embedded appearance of $R^\mu_{\lambda\sigma\tau}$ in equation (5.5) is all that is required. We conclude that

$$R^\mu_{\lambda\sigma\tau} = \frac{\partial \Gamma^\mu_{\lambda\sigma}}{\partial x^\tau} - \frac{\partial \Gamma^\mu_{\lambda\tau}}{\partial x^\sigma} + \Gamma^\mu_{\nu\tau} \Gamma^\nu_{\lambda\sigma} - \Gamma^\mu_{\nu\sigma} \Gamma^\nu_{\lambda\tau} \quad (5.6)$$

is a true tensor. It is known as the *Riemann curvature tensor*. In fact, it may be shown² that this is the only tensor that is *linear* in the second derivatives of $g_{\mu\nu}$ and contains only its first and second derivatives. The linearity in the second derivatives of the metric tensor is an important property of the curvature tensor, ultimately allowing general relativity to reduce to Newtonian gravity in the appropriate weak-field limit.

Why do we refer to this peculiar mixed tensor as the “curvature tensor?” We may begin to answer this by noting that it vanishes in ordinary flat Minkowski spacetime—simply choose Cartesian coordinates to do the calculation. Then, because $R^\mu_{\lambda\sigma\tau}$ is a tensor, if it is zero in one set of coordinates, it is zero in all. The fact that covariant derivatives commute in this case makes perfect sense. Evidently, distortions from Minkowski space are essential for a nonvanishing curvature tensor, an intuition we shall strengthen in the next section.

Exercise 5.1.1

What is the (much simpler) form of $R^\mu_{\lambda\sigma\tau}$ in local inertial coordinates? Justify any terms that are retained. It is often convenient to work in such coordinates to prove a result, and then generalise it to arbitrary coordinates using the fact that $R^\mu_{\lambda\sigma\tau}$ is a tensor.

²W72, 134.

5.2 The curvature tensor via basis vectors³

Let $\mathbf{e}_\lambda$ be a set of contravariant basis vectors, following §4.2.6. We consider the quantity

$$(\partial_\mu \partial_\nu - \partial_\nu \partial_\mu) \mathbf{e}_\lambda. \quad (5.7)$$

If we think of an n -dimensional basis vector as a collection of n mathematical functions, then equation (5.7) is identically zero, because partial derivatives commute when applied to each of the individual component functions. For example, the simple radial basis vector in spherical coordinates is

$$\mathbf{e}_r = \sin \theta \cos \phi \mathbf{e}_x + \sin \theta \sin \phi \mathbf{e}_y + \cos \theta \mathbf{e}_z,$$

and this would certainly vanish if inserted into equation (5.7) for any μ or ν . But this vanishing is, more generally, *not* compatible with equation (4.49), which defines the affine connection as an expansion coefficient in the expression for the partial derivative of $\mathbf{e}_\lambda$:

$$\partial_\mu \mathbf{e}_\lambda = \Gamma_{\mu\lambda}^\kappa \mathbf{e}_\kappa. \quad (5.8)$$

A direct application of this coefficient to equation (5.7) then directly yields

$$(\partial_\mu \partial_\nu - \partial_\nu \partial_\mu) \mathbf{e}_\lambda = [(\partial_\mu \Gamma_{\nu\lambda}^\kappa - \partial_\nu \Gamma_{\mu\lambda}^\kappa) \mathbf{e}_\kappa + (\Gamma_{\mu\rho}^\kappa \Gamma_{\nu\lambda}^\rho - \Gamma_{\nu\rho}^\kappa \Gamma_{\mu\lambda}^\rho)] \mathbf{e}_\kappa = R_{\nu\lambda\mu}^\kappa \mathbf{e}_\kappa, \quad (5.9)$$

where $R_{\nu\lambda\mu}^\kappa$ is the Riemann curvature tensor! In the previous section, we showed that this tensor emerges when one calculates the commutator of *covariant* derivatives of the components V^μ of a (contravariant) vector. By striking contrast, we have just seen that if we calculate the commutator of simple partial derivatives operating on a basis vector $\mathbf{e}_\lambda$, the curvature tensor once again emerges. How may we best understand this?

If $R_{\nu\lambda\mu}^\kappa = 0$, then the familiar rule that partial derivatives commute when applied to ordinary functions also applies to the basis vectors $\mathbf{e}_\lambda$ themselves. More generally, however, basis vectors are a different type of mathematical entity. If partial derivative commutation does hold for the basis vectors (because of a vanishing curvature tensor), the basis vectors in question may then be expressed as a linear combination of flat-space (Cartesian or Minkowskian) basis vectors, just as we did for $\mathbf{e}_r$ above. The coefficients multiplying these flat-space basis vectors are then just well-behaved mathematical functions. Indeed, we may think of this as representing a direct coordinate transformation on an intrinsically flat space. If the curvature tensor does not vanish, no such superposition of flat space vectors is possible; there is no such coordinate transformation. This is not quite a rigorous proof that $R_{\mu\nu\lambda}^\kappa = 0$ implies an underlying flat space, because we have simply asserted (intuitively) that the most general solution of a vanishing partial derivative commutator condition is one in which these derivatives must operate on well-behaved functions. But this is, in fact, true, and equation (5.9) provides a readily grasped and very plausible explanation for why the vanishing of the curvature tensor ensures that the underlying spacetime is flat.

³The results of this section are due to Callum Bell. I am grateful to him for bringing this elegant approach to my attention.

Exercise 5.2.1

Let $V = V^\lambda e_\lambda$ be a basis function expansion of a vector in terms of its contravariant components. Show that

$$(\partial_\mu \partial_\nu - \partial_\nu \partial_\mu) V = R^\kappa{}_{\nu\lambda\mu} V^\lambda e_\kappa,$$

and

$$(V_{;\mu})_{;\nu} - (V_{;\nu})_{;\mu} = R^\lambda{}_{\kappa\mu\nu} V^\kappa e_\lambda.$$

5.3 Round trip by parallel transport

We consider a vector V_λ whose covariant derivative along a curve $x(\tau)$ vanishes. Then,

$$\frac{dV_\lambda}{d\tau} = \Gamma^\mu_{\lambda\nu} \frac{dx^\nu}{d\tau} V_\mu \rightarrow dV_\lambda = \Gamma^\mu_{\lambda\nu} dx^\nu V_\mu. \quad (5.10)$$

Next, we consider a tiny round-trip journey over a closed path on which V_λ is changing by the above prescription. If we remain in the neighborhood of some point X^ρ , with coordinate x^ρ passing through X^ρ at some instant $\bar{\tau}$, so that $x^\rho(\bar{\tau}) = X^\rho$, we Taylor expand as follows:

$$\Gamma^\mu_{\lambda\nu}(x) = \Gamma^\mu_{\lambda\nu}(X) + (x^\rho - X^\rho) \frac{\partial \Gamma^\mu_{\lambda\nu}}{\partial X^\rho} + \dots \quad (5.11)$$

$$V_\mu[x(\tau)] = V_\mu(X) + dV_\mu + \dots = V_\mu(X) + (x^\rho - X^\rho) \Gamma^\sigma_{\mu\rho}(X) V_\sigma(X) + \dots, \quad (5.12)$$

where $x^\rho - X^\rho$ is identified with dx^ρ following the right side of the parallel transport equation (5.10), whence

$$\Gamma^\mu_{\lambda\nu}(x) V_\mu(x) = \Gamma^\mu_{\lambda\nu} V_\mu + (x^\rho - X^\rho) V_\sigma \left(\frac{\partial \Gamma^\sigma_{\lambda\nu}}{\partial X^\rho} + \Gamma^\sigma_{\mu\rho} \Gamma^\mu_{\lambda\nu} \right) + \dots, \quad (5.13)$$

where all quantities on the right (except x^ρ) are evaluated at the point X . Integrating

$$dV_\lambda = \Gamma^\mu_{\lambda\nu}(x) V_\mu(x) dx^\nu \quad (5.14)$$

around a tiny closed path $\oint$, and using equation (5.13), we find there is a change in the starting value ΔV_λ arising from the term linear in x^ρ given by

$$\Delta V_\lambda = \left(\frac{\partial \Gamma^\sigma_{\lambda\nu}}{\partial X^\rho} + \Gamma^\sigma_{\mu\rho} \Gamma^\mu_{\lambda\nu} \right) V_\sigma \oint x^\rho dx^\nu. \quad (5.15)$$

The integral $\oint x^\rho dx^\nu$ certainly does not vanish. (Try integrating it around a unit square in the xy -plane.) But it *is* antisymmetric in ρ and ν . (Integrate $x^\rho dx^\nu$ by parts and note that the integrated term vanishes, being an exact differential on a closed path.) That means the only part of the $\Gamma\Gamma$ and $\partial\Gamma/\partial X$ terms that survives the $\rho\nu$ summation is the part that is antisymmetric in (ρ, ν) . Since any object depending on two indices, say $A(\rho, \nu)$, can be

written as a symmetric part plus an antisymmetric part,

$$\frac{1}{2}[A(\rho, \nu) + A(\nu, \rho)] + \frac{1}{2}[A(\rho, \nu) - A(\nu, \rho)],$$

we find

$$\Delta V_\lambda = \frac{1}{2} R^\sigma_{\lambda\nu\rho} V_\sigma \oint x^\rho dx^\nu, \quad (5.16)$$

where

$$R^\sigma_{\lambda\nu\rho} = \left(\frac{\partial \Gamma^\sigma_{\lambda\nu}}{\partial X^\rho} - \frac{\partial \Gamma^\sigma_{\lambda\rho}}{\partial X^\nu} + \Gamma^\sigma_{\mu\rho} \Gamma^\mu_{\lambda\nu} - \Gamma^\sigma_{\mu\nu} \Gamma^\mu_{\lambda\rho} \right) \quad (5.17)$$

is precisely the curvature tensor. Parallel transport of a vector around a closed curve does not change the vector, unless the enclosed area has a nonvanishing curvature tensor. In fact, “the enclosed area” can be given a more intuitive interpretation if we think of integrating around a very tiny square in the $\rho\nu$ -plane. Then, the closed loop integral is just the directed area $dx^\rho dx^\nu$:

$$\Delta V_\lambda = \frac{1}{2} R^\sigma_{\lambda\nu\rho} V_\sigma dx^\rho dx^\nu. \quad (5.18)$$

The conversion of a tiny closed-loop integral to an enclosed surface area element reminds us of Stokes’s theorem, and it will not be surprising to see that this is analogous to the identity “divergence of curl equals zero.” We will make good use of this result shortly.

Exercise 5.3.1

A laboratory demonstration. Take a pencil and move it round the surface of a flat desktop without rotating the pencil. Do the same on a cylinder, something like a mailing tube. (Avoid the endcap!) Moving the pencil around a closed path, *always parallel to itself*, will in either case not change its orientation. Now, do the same on the surface of a globe. Take a small pencil, pointed poleward, and move it from the equator along the 0° meridian through Greenwich till you hit the north pole. Now, move the pencil, once again parallel to itself, down the 90°E meridian till you come to the equator. Finally, slide the pencil, once again parallel to itself, along the equator to return to the starting point at the prime meridian.

Has the pencil orientation changed from its initial one? Explain.

Exercise 5.3.2

A theoretical demonstration. Show that the parallel transport equations of a covariant vector A_μ on the surface of a sphere are

$$dA_\theta = \cot \theta A_\phi d\phi, \quad dA_\phi = \cot \theta A_\phi d\theta - \sin \theta \cos \theta A_\theta d\phi.$$

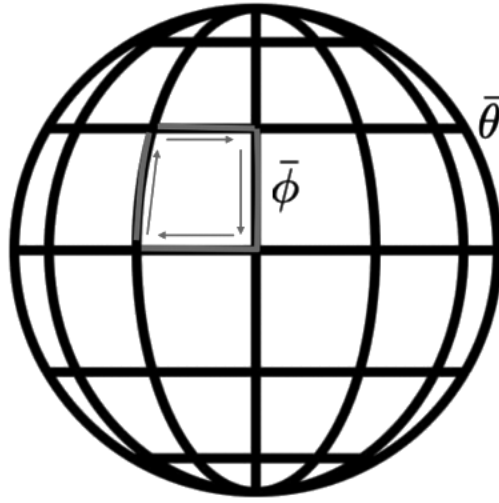


Figure 5.1 Parallel transport path for Exercise 5.3.2. Start at the bottom left corner of the gray spherical rectangle, $\theta = \pi/2$, $\phi = 0$. Move upward at constant ϕ to $\theta = \bar{\theta}$. Then rightward from $\phi = 0$ to $\phi = \bar{\phi}$ at constant θ . Return downward at constant ϕ , with θ advancing from $\bar{\theta}$ to $\pi/2$. Finally, move leftward back to the starting point at constant θ .

Begin with $A_\theta = -1$ (north pointing), $A_\phi = 0$ at $\theta = \pi/2$, $\phi = 0$. Parallel transport A_μ along the four-legged path of (1) $\phi = 0$ (fixed), θ going from $\pi/2$ to $\bar{\theta}$; (2) ϕ going from 0 to $\bar{\phi}$, $\theta = \bar{\theta}$ (fixed); (3) $\phi = \bar{\phi}$ (fixed), θ going from $\bar{\theta}$ to $\pi/2$; (4) ϕ going from $\bar{\phi}$ to $\phi = 0$, $\theta = \pi/2$ (fixed). (See fig. 5.1.) Show that, upon returning to its starting point, A_μ has become

$$A_\theta = \cos(\bar{\phi} \cos \bar{\theta}), \quad A_\phi = -\sin(\bar{\phi} \cos \bar{\theta}).$$

Verify this result qualitatively once again by direct experiment with a pencil and a globe.

Curvature, or more precisely the departure of spacetime from Minkowski structure, reveals itself through the existence of the curvature tensor $R^\sigma_{\lambda\nu\rho}$. If spacetime is Minkowski-flat, every component of the curvature tensor vanishes. An important consequence is that parallel transport around a closed loop can result in a vector or tensor not returning to its original value, if the closed loop encompasses matter (or its energy equivalent). A satellite experiment was proposed in 1961 to measure the “geodetic precession” of a gyroscope orbiting Earth caused by the effects of both spacetime curvature (which we have been examining) and the dragging of inertial frames due to Earth’s rotation (see chapter 7). Plagued by long delays, this proposal eventually evolved into a satellite known as Gravity Probe B, launched 43 years later(!), in 2004. Continuing its streak of bad luck, the experiment was beset by technical problems for several years, and the observations suffered from unexpectedly high noise levels (due to solar wind activity). Questions as to

whether systematic sources of error were understood made the data analysis controversial. A final peer-reviewed publication⁴ of the science results in 2011 verified both predictions of general relativity: the geodetic precession to an accuracy of 0.1% and the more elusive frame-dragging effect to an accuracy of 5%.

The impact of Gravity Probe B was diluted by the project's long delays. When first proposed, experimental tests of general relativity were scarce, still restricted to the classical triad of Mercury's perihelion advance, solar starlight deflection, and the gravitational redshift. Any new and potentially observable result would have been novel and worthy of detailed exploration. In the intervening half century, experimental tests of general relativity evolved tremendously. We now live in a world of meter-resolution GPS navigation, strong gravitational lensing, exquisitely sensitive Shapiro time-delays, routine detection of gravitational radiation, and direct imaging of the shadows cast by the event horizons of the black holes in our Galaxy and in M87. That said, while the demand to measure the leading-order effects of general relativistic parallel transport and frame dragging may no longer be quite so brisk, Gravity Probe B has provided the lone direct experimental verification of these two distinctive and remarkable predictions of relativistic gravity.

5.4 Geodesic deviation

The curvature tensor also appears when we calculate the difference δx^μ in geodesic space-time paths executed by two closely spaced curves, $x^\mu(t)$ and $x'^\mu(t) = x^\mu(t) + \delta x^\mu(t)$. The covariant derivative of the differential contravariant vector δx^μ is given by (cf. equation [4.86]):

$$\frac{D\delta x^\mu}{D\tau} = \frac{d\delta x^\mu}{d\tau} + \Gamma_{\nu\lambda}^\mu \frac{dx^\nu}{d\tau} \delta x^\lambda. \quad (5.19)$$

Applying the $D/D\tau$ operator to $D\delta x^\mu/D\tau$ itself then gives us

$$\frac{D^2\delta x^\mu}{D\tau^2} = \frac{d^2\delta x^\mu}{d\tau^2} + \frac{\partial\Gamma_{\nu\lambda}^\mu}{\partial x^\rho} \frac{dx^\rho}{d\tau} \frac{dx^\nu}{d\tau} \delta x^\lambda + \dots, \quad (5.20)$$

where the $\dots$ notation indicates terms with at least one power of Γ present. We will work in local inertial coordinates so that these Γ 's vanish, but we must be careful to retain the Γ -derivatives, which do not.

To evaluate $d^2\delta x^\mu/d\tau^2$, we use the fact that both $x^\mu(\tau)$ and $x'^\mu(\tau) = x^\mu(\tau) + \delta x^\mu(\tau)$ satisfy the geodesic equation. The difference between these two geodesic equations is, to first order in δx^μ ,

$$0 = \frac{d^2\delta x^\mu}{d\tau^2} + \frac{\partial\Gamma_{\nu\lambda}^\mu}{\partial x^\rho} \delta x^\rho \frac{dx^\nu}{d\tau} \frac{dx^\lambda}{d\tau} + \dots, \quad (5.21)$$

where once again $\dots$ denotes the additional terms that are proportional to Γ -factors and therefore vanish in inertial coordinates. If we now swap the ρ and λ indices in equation

⁴Everitt, C.W.F.; et al. 2011, *Phys. Rev. Lett.*, 106, 221101.

(5.21) and combine this result with equation (5.20), we obtain

$$\frac{D^2 \delta x^\mu}{D\tau^2} = \left(\frac{\partial \Gamma^\mu_{\nu\lambda}}{\partial x^\rho} - \frac{\partial \Gamma^\mu_{\nu\rho}}{\partial x^\lambda} \right) \frac{dx^\rho}{d\tau} \frac{dx^\nu}{d\tau} \delta x^\lambda + \dots = R^\mu_{\nu\lambda\rho} \frac{dx^\rho}{d\tau} \frac{dx^\nu}{d\tau} \delta x^\lambda. \quad (5.22)$$

The final equality, which is exact, follows because the surviving Γ -derivatives combine to form precisely the curvature tensor in these inertial coordinates. But this is a vector equation, and so this must be true in any other coordinate system, as well! The presence of curvature implies that nearby geodesics will deviate from one another, or, conversely, that the existence of such deviations mean that true spacetime curvature is present. The principle of equivalence tells us that while we have no way of measuring the gravitational field *locally* as we are all falling along our common geodesic, the very small, higher-order geodesic deviations can, in fact, reveal an ambient gravitational field. In the Newtonian limit, these are just the familiar tidal forces. We will see this more explicitly in the next section, but only if you work exercise 5.5.1 below.

5.5 Algebraic identities of $R^\sigma_{\nu\lambda\rho}$

5.5.1 Remembering the curvature tensor formula

It is helpful to have a mnemonic for generating the curvature tensor. The hard part is keeping track of the indices. Recall that the tensor itself is just a sum of derivatives of Γ , and quadratic products of Γ . That combination is easy to remember, since the curvature tensor has “dimensions” of $1/x^2$, where x represents a coordinate, and $\Gamma \sim 1/x$. To remember the coordinate juggling of R^a_{bcd} , we start with

$$\frac{\partial \Gamma^a_{bc}}{\partial x^d} + \Gamma^*_{bc} \Gamma^a_{d*},$$

where the first $abcd$ ordering in the Γ -derivative is not difficult to remember, since it follows much the same downward placement in R^a_{bcd} . For the $\Gamma\Gamma$ term, begin and end with a dummy variable $*$, filling in $abcd$ starting from the upper left $*$, continuing across with a , and then to the bottom row with $bcd*$. To finish, just subtract the expression you have with c and d reversed, which ensures the proper antisymmetric behavior of these indices. You arrive at

$$R^a_{bcd} = \frac{\partial \Gamma^a_{bc}}{\partial x^d} - \frac{\partial \Gamma^a_{bd}}{\partial x^c} + \Gamma^*_{bc} \Gamma^a_{d*} - \Gamma^*_{bd} \Gamma^a_{c*}, \quad (5.23)$$

the correct form of the curvature tensor.

5.5.2 $R_{\lambda\mu\nu\kappa}$: Fully covariant form

The fully covariant form of the curvature tensor can be written in a useful form that involves only second-order derivatives of $g_{\mu\nu}$ and products of Γ s, with no other explicit Γ partial derivatives. The second-order g -derivatives, which are *linear* terms, will be our point of

contact with Newtonian theory (as well as with weak gravitational radiation) from the full field equations. But we do first have a bit of heavy weather ahead!

We start with

$$R_{\lambda\mu\nu\kappa} = g_{\lambda\sigma} R^\sigma_{\mu\nu\kappa}, \quad (5.24)$$

or

$$R_{\lambda\mu\nu\kappa} = g_{\lambda\sigma} \left[\frac{\partial \Gamma^\sigma_{\mu\nu}}{\partial x^\kappa} - \frac{\partial \Gamma^\sigma_{\mu\kappa}}{\partial x^\nu} + \Gamma^\eta_{\mu\nu} \Gamma^\sigma_{\kappa\eta} - \Gamma^\eta_{\mu\kappa} \Gamma^\sigma_{\nu\eta} \right]. \quad (5.25)$$

From the definition of the affine connection, equation (3.21), the right side of equation (5.25) is

$$\begin{aligned} & \frac{g_{\lambda\sigma}}{2} \frac{\partial}{\partial x^\kappa} \left[g^{\sigma\rho} \left(\frac{\partial g_{\rho\mu}}{\partial x^\nu} + \frac{\partial g_{\rho\nu}}{\partial x^\mu} - \frac{\partial g_{\mu\nu}}{\partial x^\rho} \right) \right] - \frac{g_{\lambda\sigma}}{2} \frac{\partial}{\partial x^\nu} \left[g^{\sigma\rho} \left(\frac{\partial g_{\rho\mu}}{\partial x^\kappa} + \frac{\partial g_{\rho\kappa}}{\partial x^\mu} - \frac{\partial g_{\mu\kappa}}{\partial x^\rho} \right) \right] \\ & + g_{\lambda\sigma} (\Gamma^\eta_{\mu\nu} \Gamma^\sigma_{\kappa\eta} - \Gamma^\eta_{\mu\kappa} \Gamma^\sigma_{\nu\eta}). \end{aligned} \quad (5.26)$$

The x^κ and x^ν partial derivatives will operate on the $g^{\sigma\rho}$ term and the g -derivative terms. Let us begin with the second group, the twice-differentiated g -derivatives, as it is simpler. With $g_{\lambda\sigma} g^{\sigma\rho} = \delta^\rho_\lambda$, the surviving four terms of this group, all linear in the second-order g -derivatives, are

$$\frac{1}{2} \left(\frac{\partial^2 g_{\lambda\nu}}{\partial x^\kappa \partial x^\mu} - \frac{\partial^2 g_{\mu\nu}}{\partial x^\kappa \partial x^\lambda} - \frac{\partial^2 g_{\lambda\kappa}}{\partial x^\nu \partial x^\mu} + \frac{\partial^2 g_{\mu\kappa}}{\partial x^\nu \partial x^\lambda} \right). \quad (5.27)$$

If you can sense the beginnings of the classical wave equation lurking in these linear second-order derivatives, which are the leading terms when $g_{\mu\nu}$ departs only a little from $\eta_{\mu\nu}$, then you are very much on the right track.

Next, we have the terms proportional to the x^κ - and x^ν -derivatives of $g^{\sigma\rho}$, which certainly do not vanish in general. But the covariant derivative of the metric tensor $g_{\lambda\sigma}$ *does* vanish, so we invoke this sleight-of-hand integration by parts:

$$g_{\lambda\sigma} \frac{\partial g^{\sigma\rho}}{\partial x^\kappa} = -g^{\sigma\rho} \frac{\partial g_{\lambda\sigma}}{\partial x^\kappa} = -g^{\sigma\rho} (\Gamma^\eta_{\kappa\lambda} g_{\eta\sigma} + \Gamma^\eta_{\kappa\sigma} g_{\eta\lambda}), \quad (5.28)$$

where in the final equality, we used equation (4.36). When we bring $g^{\sigma\rho}$ out from the partial derivative, it recombines with the first-order g -derivatives to form affine connections once again. All the remaining terms of $R_{\lambda\mu\nu\kappa}$ from equation (5.26) are now of the form $g\Gamma\Gamma$:

$$- (\Gamma^\eta_{\kappa\lambda} g_{\eta\sigma} + \Gamma^\eta_{\kappa\sigma} g_{\eta\lambda}) \Gamma^\sigma_{\mu\nu} + (\Gamma^\eta_{\nu\lambda} g_{\eta\sigma} + \Gamma^\eta_{\nu\sigma} g_{\eta\lambda}) \Gamma^\sigma_{\mu\kappa} + g_{\lambda\sigma} (\Gamma^\eta_{\mu\nu} \Gamma^\sigma_{\kappa\eta} - \Gamma^\eta_{\mu\kappa} \Gamma^\sigma_{\nu\eta}). \quad (5.29)$$

It is not obvious at first, but using some index agility, you should be able to see four of these six $g\Gamma\Gamma$ terms cancel out—the second group with the fifth, the fourth group with the sixth—leaving only the first and third terms:

$$g_{\eta\sigma} (\Gamma^\eta_{\nu\lambda} \Gamma^\sigma_{\mu\kappa} - \Gamma^\eta_{\kappa\lambda} \Gamma^\sigma_{\mu\nu}). \quad (5.30)$$

Adding together the terms in equations (5.27) and (5.30), we arrive at

$$R_{\lambda\mu\nu\kappa} = \frac{1}{2} \left(\frac{\partial^2 g_{\lambda\nu}}{\partial x^\kappa \partial x^\mu} - \frac{\partial^2 g_{\mu\nu}}{\partial x^\kappa \partial x^\lambda} - \frac{\partial^2 g_{\lambda\kappa}}{\partial x^\nu \partial x^\mu} + \frac{\partial^2 g_{\mu\kappa}}{\partial x^\nu \partial x^\lambda} \right) + g_{\eta\sigma} (\Gamma_{\nu\lambda}^\eta \Gamma_{\mu\kappa}^\sigma - \Gamma_{\kappa\lambda}^\eta \Gamma_{\mu\nu}^\sigma). \quad (5.31)$$

Exercise 5.5.1

What is $R_{\lambda\mu\nu\kappa}$ in local inertial coordinates? Why is this the same form as emerges in the Newtonian limit? Show that for a static potential Φ , the equation of geodesic deviation (5.22) for a Cartesian spatial component in the Newtonian limit reduces to

$$\frac{d^2 \delta x^i}{dt^2} = - \frac{\partial^2 \phi}{\partial x^i \partial x^j} \delta x^j.$$

(Hint: Note that you may raise indices with $\eta^{\alpha\beta}$ [why?], and only the $R_{\lambda 0 \nu 0}$ term survives from $R_{\lambda\mu\nu\kappa}$ [why?].) Give a physical interpretation of this equation as a tidal-force manifestation.

Note the following important symmetry properties for the indices of $R_{\lambda\mu\nu\kappa}$. Because each of these identities may be expressed as a vanishing tensor equation (left side minus right side equals zero), they may be established generally by choosing any coordinates we like. We therefore choose a simple, locally inertial frame in which the Γ vanish. These results may then be verified just from the terms linear in the g -derivatives in equation (5.31):

$$R_{\lambda\mu\nu\kappa} = R_{\nu\kappa\lambda\mu} \quad (\text{symmetry}), \quad (5.32)$$

$$R_{\lambda\mu\nu\kappa} = -R_{\mu\lambda\nu\kappa} = -R_{\lambda\mu\kappa\nu} = R_{\mu\lambda\kappa\nu} \quad (\text{antisymmetry}), \quad (5.33)$$

$$R_{\lambda\mu\nu\kappa} + R_{\lambda\kappa\mu\nu} + R_{\lambda\nu\kappa\mu} = 0 \quad (\text{cyclic}). \quad (5.34)$$

5.5.3 How many independent components does the curvature tensor have?

To count the number of independent components of $R_{\lambda\mu\nu\kappa}$ in n dimensions, we start with the simple observation that an antisymmetric tensor $A_{\mu\nu}$ has $n(n-1)/2$ independent components in n dimensions: there are n choices for μ , only $n-1$ choices for ν (since the tensor vanishes if $\mu = \nu$), and we must divide by 2, since $A_{\mu\nu} = -A_{\nu\mu}$. There are $n(n+1)/2$ independent components of a symmetric tensor, say $S_{\mu\nu}$, since we simply add back in the n diagonal elements which are now present but that we neglected in the component counting of $A_{\mu\nu}$. Next, we mentally group the indices of the covariant form of the curvature tensor as $R_{(\lambda\mu)(\nu\kappa)}$. By the antisymmetry properties of equation (5.33), there are $n(n-1)/2$ possible choices for each $\lambda\mu$ or $\kappa\nu$ pair that will lead to distinct $R_{\lambda\mu\nu\kappa}$ components. With $(\lambda\mu)$ or $(\kappa\nu)$ each viewed as a sort of mega-index unto itself, the symmetry of equation (5.32) now

tells us there are

$$\frac{1}{2} \left[\frac{n(n-1)}{2} \right] \left[\frac{n(n-1)}{2} + 1 \right] \equiv N \quad (5.35)$$

ways to list the components of $R_{(\lambda\mu)(\nu\kappa)}$.

But we are not yet finished, because we have not yet taken into account equation (5.34), the cyclic sum. Note that equations (5.32) and (5.33) render this sum completely antisymmetric if we interchange any two indices. So, we need count only those combinations where the four indices are different, i.e., the number of distinct combinations of four elements taken from a set of n . In $n = 4$ dimensions this is clearly only one constraint. In $n > 4$ dimensions, we have exactly one equation for each unique combination of $\lambda\mu\nu\kappa$ with no index repeats. This is just the number of combinations of four items, taken from a larger set of n items with distinct members, or

$$\frac{n!}{(n-4)!4!} = \frac{1}{24} n(n-1)(n-2)(n-3). \quad (5.36)$$

The total number of independent components of $R_{\lambda\mu\nu\kappa}$ is therefore

$$N - \frac{1}{24} n(n-1)(n-2)(n-3) = \frac{1}{12} n^2(n^2 - 1). \quad (5.37)$$

In one dimension there are zero components: R_{1111} vanishes by antisymmetry. Is a curved line really not curved? Indeed, it is not, because the casual use of the word “curved” is not the same as the mathematical sense of a distortion from Euclidean (or Minkowski) spatial structure. You can use a one-dimensional piece of string to measure distances accurately along flat or curved surfaces equally well. There is simply no escape from a Euclidean metric in one dimension; it is all just a change of variable: the metric $f(x)dx \equiv dX$, where $X = \int f(x) dx$.

In two dimensions, there is one component of the curvature tensor, R_{1212} . This was first identified by Gauss in his study of the mathematics of curved surfaces, before he handed the problem in its full generality over to Riemann.

In three dimensions, there are six components, but we will see in §5.9 that this is not enough for the existence of a gravitational field in empty space. Only with at least four dimensions, with its 20 different components of the curvature tensor, do we have enough freedom to support a gravitational field from an external source (a tidal force actually) in empty space.

5.6 The Ricci tensor

The Ricci tensor is the curvature tensor contracted on its (raised) first and third indices, R^a_{bad} . In terms of the covariant curvature tensor, the Ricci tensor $R_{\mu\kappa}$ is given by

$$R_{\mu\kappa} = g^{\lambda\nu} R_{\lambda\mu\nu\kappa} = g^{\lambda\nu} R_{\nu\kappa\lambda\mu} = R^\lambda_{\kappa\lambda\mu} = R_{\kappa\mu}, \quad (5.38)$$

so that the Ricci tensor is symmetric. From equation (5.6), we have explicitly

$$R_{\mu\kappa} = \frac{\partial \Gamma_{\mu\lambda}^{\lambda}}{\partial x^{\kappa}} - \frac{\partial \Gamma_{\mu\kappa}^{\lambda}}{\partial x^{\lambda}} + \Gamma_{\mu\lambda}^{\eta} \Gamma_{\kappa\eta}^{\lambda} - \Gamma_{\mu\kappa}^{\eta} \Gamma_{\lambda\eta}^{\lambda}. \quad (5.39)$$

The Ricci tensor is of fundamental importance in general relativity. Indeed, we shall very soon see that $R_{\mu\nu} = 0$ is Einstein's Laplace equation. There is enough information here to calculate the deflection of light by a gravitating body, the advance of a planet's orbital perihelion, or the spacetime geometry of a black hole. What is tricky is to guess the general relativistic version of the *Poisson* equation, and no, it is *not* $R_{\mu\nu}$ proportional to the stress-energy tensor $T_{\mu\nu}$. (That wouldn't be very tricky then, would it?) Notice that while $R^{\lambda}_{\mu\nu\kappa} = 0$ implies that the Ricci tensor vanishes, the converse does not follow: $R_{\mu\nu} = 0$ does not necessarily mean that the full curvature tensor (covariant or otherwise) vanishes. If it did, there would never be gravitational fields in empty space. The vanishing of the Ricci tensor means only that certain linear combinations of $R^{\lambda}_{\mu\nu\kappa}$ vanish.

Exercise 5.6.1

The expression for $R_{\mu\kappa}$ given on page (xiv) has a somewhat different form of the Ricci tensor from the one above, equation (5.39). With $g = |\det g_{\mu\nu}|$, this is

$$R_{\mu\kappa} = \frac{1}{2} \frac{\partial^2 \ln g}{\partial x^{\kappa} \partial x^{\mu}} - \frac{1}{\sqrt{g}} \frac{\partial \sqrt{g} \Gamma_{\mu\kappa}^{\lambda}}{\partial x^{\lambda}} + \Gamma_{\mu\lambda}^{\eta} \Gamma_{\kappa\eta}^{\lambda}.$$

Show that the two expressions are equivalent.

Next, show that

$$R_{\mu\kappa} = -g^{\lambda\nu} R_{\mu\lambda\nu\kappa} = -g^{\lambda\nu} R_{\lambda\mu\kappa\nu} = g^{\lambda\nu} R_{\mu\lambda\kappa\nu}$$

and that $g^{\lambda\mu} R_{\lambda\mu\nu\kappa} = g^{\nu\kappa} R_{\lambda\mu\nu\kappa} = 0$. Why does this mean that $R_{\mu\kappa}$ is the only second-rank covariant tensor that can be formed from contracting $R_{\lambda\mu\nu\kappa}$?

We are not quite through contracting. We may also form the Ricci (or curvature) scalar

$$R \equiv R^{\mu}_{\mu}, \quad (5.40)$$

the trace of the Ricci tensor, and another fundamentally important quantity in general relativity.

Exercise 5.6.2

The curvature scalar is unique. Prove that

$$R = g^{\nu\lambda} g^{\mu\kappa} R_{\lambda\mu\nu\kappa} = -g^{\nu\lambda} g^{\mu\kappa} R_{\mu\lambda\nu\kappa}$$

and that

$$g^{\lambda\mu}g^{\nu\kappa}R_{\lambda\mu\nu\kappa}=0.$$

Justify the title of this exercise.

Finally, it will be of interest later for a variational calculation to have at hand a formula for $\delta R_{\mu\kappa}$, the change in the Ricci tensor due to a change $\delta g_{\mu\nu}$ in the metric tensor, namely,

$$\delta R_{\mu\kappa} = (\delta\Gamma_{\mu\lambda}^{\lambda})_{;\kappa} - (\delta\Gamma_{\mu\kappa}^{\lambda})_{;\lambda}, \quad (5.41)$$

where the $\delta\Gamma$ are the corresponding changes in the affine connections. The easiest way to verify this identity is to note that in local inertial coordinates (with vanishing affine connection), we need retain only the first two partial derivative terms on the right side of equation (5.39). The result for general coordinates then follows by simply turning the partial derivatives into covariant derivatives, since both $\delta R_{\mu\kappa}$ and $\delta\Gamma_{\mu\nu}^{\lambda}$ are true tensors. Equation (5.41) is known as Palatini's identity. A particularly useful form of the identity is to contract (5.41) with $g^{\mu\kappa}$:

$$\begin{aligned} g^{\mu\kappa}\delta R_{\mu\kappa} &= (g^{\mu\kappa}\delta\Gamma_{\mu\lambda}^{\lambda})_{;\kappa} - (g^{\mu\kappa}\delta\Gamma_{\mu\kappa}^{\lambda})_{;\lambda} \\ &= \frac{1}{\sqrt{g}}\frac{\partial}{\partial x^{\kappa}}\left(\sqrt{g}g^{\mu\kappa}\delta\Gamma_{\mu\lambda}^{\lambda}\right) - \frac{1}{\sqrt{g}}\frac{\partial}{\partial x^{\lambda}}\left(\sqrt{g}g^{\mu\kappa}\delta\Gamma_{\mu\kappa}^{\lambda}\right) \\ &= \frac{1}{\sqrt{g}}\frac{\partial}{\partial x^{\kappa}}\left[\sqrt{g}\left(g^{\mu\kappa}\delta\Gamma_{\mu\lambda}^{\lambda} - g^{\mu\lambda}\delta\Gamma_{\mu\lambda}^{\kappa}\right)\right]. \end{aligned} \quad (5.42)$$

Note the use of the *single* divergence operator on the right.

5.7 Curvature and Newtonian gravity

That curvature should be at the heart of our understanding of gravity is perhaps not too surprising if we examine carefully the Newtonian Poisson equation,

$$\nabla^2\Phi = 4\pi G\rho, \quad (5.43)$$

where Φ is the usual gravitational potential, and ρ the mass density. Second derivatives of functions are generally associated with literal curvature. If you have an old-fashioned-enough book on fluid dynamics available,⁵ you will come across a formula that relates the ∇^2 operator directly to the physical curvature of a standard 2D surface:

$$\frac{\partial^2\zeta}{\partial x^2} + \frac{\partial^2\zeta}{\partial y^2} = \frac{1}{R_1} + \frac{1}{R_2}, \quad (5.44)$$

⁵For example, Batchelor, G. K. 2000, *An Introduction to Fluid Dynamics*.

where ζ is the (assumed small) displacement of the surface from planar, and the R 's are the two “principal radii of curvature.” (Pick a surface point and think of fitting as smoothly as possible, into the deformed surface region, arcs of two circles at right angles to each other. The R 's in the above formula are the radii of these circles.) Thus, the Laplacian operator is, on its own, quite literally a measure of spatial curvature, at least in two dimensions. Moreover, if we have a surface membrane with a pressure discontinuity $\Delta P(x, y)$ across it, then the displacement field satisfies the dynamic force balance (Young-Laplace equation)

$$\gamma \left(\frac{\partial^2 \zeta}{\partial x^2} + \frac{\partial^2 \zeta}{\partial y^2} \right) = \Delta P, \quad (5.45)$$

where γ is the assumed constant surface tension of the membrane (dimensions of energy per area). This is identical to the two-dimensional Poisson equation. So, at least in a two-dimensional world, we can think of the Newtonian potential Φ as the displacement of a surface membrane, and ρ as the applied pressure difference. In this analogy, that makes $1/(4\pi G)$ equal to the surface tension! Finally, if we were to construct such a membrane in a constant vertical gravitational field $-g$, then, to leading order, the horizontal x - and y -accelerations of a test mass on the distorted surface would be given by $-g(\partial_x \zeta, \partial_y \zeta)$, i.e., the gradients of ζ , our analog potential.

The origins of Riemannian geometry began with Gauss trying to describe the mathematics of two-dimensional surfaces with arbitrary curvature, and, as we have noted, Riemann then worked out the full structure of curvature in complete generality. It is of some interest, therefore, to see a form of true geometric curvature already built into two-dimensional Newtonian gravity at a fundamental level.

5.8 The Bianchi identities

The fully covariant curvature tensor obeys a very important set of differential identities, analogous to $\text{div}(\text{curl})=0$, namely, the Bianchi identities. We shall prove the general Bianchi identities in our preferred local inertial coordinates with $\Gamma = 0$, and since we will be showing that a tensor is zero in these coordinates, it must be zero in all coordinate systems. In $\Gamma = 0$ coordinates,

$$R_{\lambda\mu\nu\kappa;\eta} = \frac{1}{2} \frac{\partial}{\partial x^\eta} \left(\frac{\partial^2 g_{\lambda\nu}}{\partial x^\kappa \partial x^\mu} - \frac{\partial^2 g_{\mu\nu}}{\partial x^\kappa \partial x^\lambda} - \frac{\partial^2 g_{\lambda\kappa}}{\partial x^\mu \partial x^\nu} + \frac{\partial^2 g_{\mu\kappa}}{\partial x^\nu \partial x^\lambda} \right). \quad (5.46)$$

The Bianchi identities follow from cycling κ goes to ν , ν goes to η , η goes to κ . We leave λ and μ alone and repeat. We then add the original $R_{\lambda\mu\nu\kappa;\eta}$ and the two cycled expressions together, which gives

$$R_{\lambda\mu\nu\kappa;\eta} + R_{\lambda\mu\eta\nu;\kappa} + R_{\lambda\mu\kappa\eta;\nu} = 0. \quad (5.47)$$

An easy way to check the bookkeeping on this process is to focus just upon the g 's: once you've picked a particular value of $\partial^2 g_{ab}$ in the numerator, the other ∂x^c indices downstairs are unambiguous, since as coordinate derivatives, their order is immaterial. The first term in equation (5.47) is then as shown: $(g_{\lambda\nu}, -g_{\mu\nu}, -g_{\lambda\kappa}, g_{\mu\kappa})$. Cycle to get the second group for the second Bianchi term, $(g_{\lambda\eta}, -g_{\mu\eta}, -g_{\lambda\nu}, g_{\mu\nu})$. Cycle once more to get the final term,

which is then $(g_{\lambda\kappa}, -g_{\mu\kappa}, -g_{\lambda\eta}, g_{\mu\eta})$. Notice that every g_{ab} has its opposite when you add up all these terms, so the sum is zero.

We would like to get equation (5.47) into the form of a *single* vanishing covariant tensor divergence, for reasons that will soon become very clear. Toward this goal, we contract λ with ν , remembering the symmetries in equation (5.33). (For example, in the second term on the left side of equation [5.47], we swap ν and η before contracting, changing the sign.) We find

$$R_{\mu\kappa;\eta} - R_{\mu\eta;\kappa} + R_{\mu\kappa\eta;\nu}^\nu = 0. \quad (5.48)$$

Next, we contract μ with κ :

$$R_{;\eta} - R_{\eta;\mu}^\mu - R_{\eta;\nu}^\nu = 0. \quad (5.49)$$

To show more explicitly how to get the last term on the left, we first set things up with

$$R_{\mu\kappa\eta;\nu}^\nu = g^{\nu\sigma} R_{\sigma\mu\kappa\eta;\nu} = -g^{\nu\sigma} R_{\mu\sigma\kappa\eta;\nu}.$$

Now, it is easy to raise μ and contract with κ :

$$-g^{\nu\sigma} R_{\sigma\mu\eta;\nu}^\mu = -g^{\nu\sigma} R_{\sigma\eta;\nu}^\mu = -R_{\eta;\nu}^\nu.$$

When we clean things up, our contracted identity (5.49) becomes

$$(\delta_{\eta}^{\mu} R - 2R_{\eta}^{\mu})_{;\mu} = 0. \quad (5.50)$$

Raising η (we are allowed, of course, to bring $g^{\nu\eta}$ *inside* the covariant derivative to do this—why?), and dividing by -2 puts this identity into its classic “zero-divergence” form:

$$\left(R^{\mu\nu} - g^{\mu\nu} \frac{R}{2} \right)_{;\mu} = 0. \quad (5.51)$$

The generic tensor combination $A^{\mu\nu} - g^{\mu\nu} A/2$ will appear repeatedly in our study of gravitational radiation. Equation (5.51) is sometimes known as the *contracted* Bianchi identities.

Einstein unfortunately did not know equation (5.51) when he was struggling mightily with the final formulation of his theory. But to be fair, neither did most mathematicians! The general form of the identities was actually first discovered by the German mathematician A. Voss in 1880, then independently in 1889 by Ricci. These results were then quickly forgotten, even, it seems, by Ricci himself! Bianchi rediscovered them on his own in 1902, but they were *still* not widely known in the mathematics community in 1915. This was a pity, because the Bianchi identities streamline the derivation of the gravitational field equations. They are referred to as the “royal road to the Gravitational Field Equations” by Einstein’s biographer, A. Pais. It seems to have been the great mathematician H. Weyl who in 1917 first recognized the importance of the Bianchi identities for relativity, yet the particular derivation we have followed was not formulated until 1922, by Harward. Finally, for the important

contracted form of the identities, we give a very simple, direct derivation in §5.11. It is truly a checkered legacy!

The reason for the identities' importance is precisely analogous to Maxwell's penetrating understanding of the tight mathematical restrictions the curl operator imposes on the type of field generated from the source vector. Why does the displacement current need to be added to the equation $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$? Because taking the divergence of this equation in this form gives zero identically on the left—the divergence of the curl is zero—so the right-hand source term must also have a vanishing divergence. In other words, it must become a statement of some sort of *physical* conservation law. Maxwell needed, and invoked, a second physical “displacement current,” $(1/c^2)\partial \mathbf{E}/\partial t$, adding it to the right side of the equation, because $\nabla \cdot \mathbf{J}$ is manifestly *not* always zero if charges are accumulating or dispersing. The ensuing physical conservation law corresponds to the conservation of electric charge, now built directly into the fundamental formulation of Maxwell's equations, and is ultimately related to the gauge invariance of the final theory. Here, we shall use the Bianchi identities as the analog (and it really is a precise mathematical analog) of “the divergence of the curl is zero,” a *geometric* constraint that ensures that conservation of the stress-energy tensor is automatically built into the fundamental formulation of Einstein's field equations, just as charge conservation is built into the underlying structure of Maxwell's field equations. What is good for Maxwell is good for Einstein.

5.9 The Einstein field equations

We will now apply the principle of general covariance to the gravitational field itself. What is the relativistic analog of $\nabla^2 \Phi = 4\pi G \rho$? By now we have built up a sufficiently strong mathematical arsenal from Riemannian geometry and Ricci calculus to be able to give a satisfactory answer to this question.

We know that we must work with vectors and tensors to maintain general covariance, and that the Newton-Poisson source, ρ , is only one component of a more general stress-energy tensor $T_{\mu\nu}$ (in its covariant form) in relativity. We expect, therefore, that the gravitational field equations will take the form

$$G_{\mu\nu} = CT_{\mu\nu}, \quad (5.52)$$

where C is a constant, and $G_{\mu\nu}$ is some tensor that comprises $g_{\mu\nu}$ and its second derivatives, or quadratic products of the first derivatives of $g_{\mu\nu}$. We guess this since (i) we know that in the Newtonian limit the derivative of $g_{00} \simeq -1 - 2\Phi/c^2$ enters directly into the dynamics; (ii) in the same limit, we need to recover the Poisson equation, which involves a Laplacian of g_{00} ; and (iii) we are assuming that our theory of gravity does not change its behavior with length scale: it has no characteristic length associated with it where the field properties change fundamentally. In other words, we need exclusively “ $1/r^2$ ” scaling on the left side.

The last condition may strike the reader as a bit too restrictive. In fact, we now know that this is actually wrong. It is wrong when applied to the Universe as a whole. But it is the simplest assumption we can make that will satisfy all the basic requirements of a good theory. We shall come back to the general relativity updates once we have installed version GR1.0.

Next, we know that the stress-energy tensor is conserved in the sense of $T^{\mu\nu}_{;\mu} = 0$. We also know from our work with the Bianchi identities of the previous section that this condition

will automatically be satisfied if we take $G_{\mu\nu}$ to be proportional to the particular linear combination

$$G_{\mu\nu} \propto R_{\mu\nu} - \frac{g_{\mu\nu}R}{2}.$$

(As always, there is no difficulty shifting indices up or down as considerations demand: the index shifters $g_{\mu\nu}$ and $g^{\mu\nu}$ all have vanishing covariant derivatives.) We have therefore determined the simplest form of the field equations of gravity up to an overall normalization constant C :

$$R_{\mu\nu} - \frac{g_{\mu\nu}R}{2} = CT_{\mu\nu}. \quad (5.53)$$

The final step is to determine C by recovering the Newtonian limit. In this limit, $T_{\mu\nu}$ is dominated by T_{00} , and $g_{\mu\nu}$ can be replaced by $\eta_{\alpha\beta}$ when shifting indices. The leading-order derivative of $g_{\mu\nu}$ that enters into the field equations comes from

$$g_{00} \simeq -1 - \frac{2\Phi}{c^2},$$

where Φ is the usual Newtonian potential. In what follows, we use our usual convention that i, j, k indicate spatial indices, and 0 is always reserved for time.

The trace of equation (5.53) reads (raise μ , contract with ν):

$$R - \frac{4 \times R}{2} = -R = CT. \quad (5.54)$$

Substituting this equality for R back into the original equation leads to

$$R_{\mu\nu} = C \left(T_{\mu\nu} - \frac{g_{\mu\nu}T}{2} \right) \equiv CS_{\mu\nu}, \quad (5.55)$$

which defines the so-called source function (a tensor, of course), a convenient grouping we shall henceforth be frequently using:

$$S_{\mu\nu} = T_{\mu\nu} - g_{\mu\nu}T/2. \quad (5.56)$$

The 00 component of equation (5.55) is

$$R_{00} = C \left(T_{00} - \frac{g_{00}T}{2} \right). \quad (5.57)$$

In the Newtonian limit, the trace $T \equiv T^\mu_\mu$ is dominated by the 0 term, T^0_0 , and we raise and lower the indices by using the $\eta_{\mu\nu}$ weak field limit of $g_{\mu\nu}$. In this static, weak field limit:

$$R_{00} = C \left(T_{00} - \frac{\eta_{00}T^0_0}{2} \right) = C \left(T_{00} - \frac{T_{00}}{2} \right) = C \frac{T_{00}}{2} = C \frac{\rho c^2}{2}, \quad (5.58)$$

where ρ is the Newtonian mass density. With R_{00} given by

$$R_{00} = R^\nu_{0\nu 0} = \eta^{\lambda\nu} R_{\lambda 0\nu 0}, \quad (5.59)$$

we require here only the linear part of $R_{\lambda\mu\nu\kappa}$:

$$R_{\lambda\mu\nu\kappa} = \frac{1}{2} \left(\frac{\partial^2 g_{\lambda\nu}}{\partial x^\kappa \partial x^\mu} - \frac{\partial^2 g_{\mu\nu}}{\partial x^\kappa \partial x^\lambda} - \frac{\partial^2 g_{\lambda\kappa}}{\partial x^\nu \partial x^\mu} + \frac{\partial^2 g_{\mu\kappa}}{\partial x^\nu \partial x^\lambda} \right). \quad (5.60)$$

In the static limit with $\mu = \kappa = 0$, only the final term on the right side of this equation survives:

$$R_{\lambda 0\nu 0} = \frac{1}{2} \frac{\partial^2 g_{00}}{\partial x^\nu \partial x^\lambda}. \quad (5.61)$$

Finally, combining this result with equation (5.58), we obtain

$$R_{00} = \eta^{\lambda\nu} R_{\lambda 0\nu 0} = \frac{1}{2} \eta^{\lambda\nu} \frac{\partial^2 g_{00}}{\partial x^\lambda \partial x^\nu} = \frac{1}{2} \nabla^2 g_{00} = -\frac{1}{c^2} \nabla^2 \Phi = \frac{C\rho c^2}{2}, \quad (5.62)$$

which happily agrees with the Poisson equation if $C = -8\pi G/c^4$. Hello, Isaac Newton. As Einstein himself put it: “No fairer destiny could be allotted to any physical theory, than that it should of itself point out the way to the introduction of a more comprehensive theory, in which it lives on as a limiting case.”⁶ We therefore arrive at the *Einstein field equations*:

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = -\frac{8\pi G}{c^4} T_{\mu\nu}. \quad (5.63)$$

The field equations first appeared in Einstein’s notes on November 25, 1915, after an inadvertent competition with the mathematician David Hilbert, triggered by an Einstein colloquium at Göttingen. (Talk about being scooped! Hilbert actually derived the field equations first, by a variational method, but rightly insisted on giving Einstein full credit for the physical theory. Incidentally, despite his prolific mathematical knowledge, Hilbert, in common with Einstein, did not know the Bianchi identities.)

It is also of interest to exhibit these equations explicitly in source function form;

$$R_{\mu\nu} = -\frac{8\pi G}{c^4} S_{\mu\nu}, \quad (5.64)$$

where $S_{\mu\nu}$ is given by equation (5.56), a “Bianchified form” of the stress tensor. In vacuo, the field equations then reduce to the analog of the Laplace equation:

$$R_{\mu\nu} = 0. \quad (5.65)$$

⁶Einstein, A.; Gutfreund, H.; & Renn, J. 2015, *Relativity: The Special and General Theory—100th Anniversary Edition* (Princeton University Press: Princeton)

It is useful, at this point, to come back to a point raised in §5.5.3, namely, that in a spacetime with three dimensions there can be no gravity, because there can be no nonvanishing curvature components. To see why this is so, recall that there are six algebraically independent components of both $R_{\lambda\mu\nu\kappa}$ and $R_{\mu\kappa}$. This means that in empty space, the relation

$$g^{\lambda\nu}R_{\lambda\mu\nu\kappa} = R_{\mu\kappa} = 0 \quad (5.66)$$

comprises six equations for the six covariant curvature tensor components. If these six independent quantities $R_{\lambda\mu\nu\kappa}$ satisfy six linear homogeneous equations, in general, the solution must be that these quantities all vanish. The space must then be flat. Curvature in gravitational theories requires at least four dimensions.

Finally, let us return to a very important point that was also raised earlier. If we allow the possibility that gravity *could* change its form on different scales of spacetime, it is always possible to add a term of the form $\Lambda g_{\mu\nu}$ to $G_{\mu\nu}$, where Λ is a constant, without violating the conservation of $T_{\mu\nu}$ condition. This is because the covariant derivatives of $g_{\mu\nu}$ vanish identically, so the conservation requirement of $T_{\mu\nu}$ (vanishing covariant divergence) is unaffected. This first emerged when Einstein, pursuing the consequences of his field equations for cosmology, realized they did not produce a static universe. This is bad, he thought, and Nature is not bad. Everyone knows that the Universe is, on its largest scales, static. How else could it be? So he introduced what he thought would be a static offset, adding a positive Λ term to the right side of the field equations as a coefficient to the metric tensor $g_{\mu\nu}$,

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = -\frac{8\pi G}{c^4}T_{\mu\nu} + \Lambda g_{\mu\nu}. \quad (5.67)$$

He dubbed Λ the *cosmological constant*. Had Einstein left well enough alone and believed in the simplest form of his equations, he could have made a spectacular prediction: the Universe is *not* a static stage. It is dynamic, a player in its own game, and must be either expanding or contracting. With the historical discovery of an expanding universe by Edwin Hubble in 1929, Einstein retracted the Λ term, calling it “the biggest mistake of my life.”

How big a mistake? Surprise—we now know that this term, or something very much like it, is, in fact, present on the largest cosmological scales, and on these scales it is no small effect. It mimics, and may well be, the energy density of the vacuum itself. The term is measured to be approximately 70% of the effective energy density in the Universe, and it certainly doesn’t make the Universe static! It is to be strongly emphasized that Λ need be taken into account only on these largest scales, scales over which the locally much greater density enhancements (such as a single galaxy) are effectively lowered by volume averaging. But gravity does indeed behave differently on different length scales. Over the largest scales of the Universe, the cosmological constant, or vacuum energy, term is important. Otherwise, Λ is completely negligible. It has been noted that the so-called biggest mistake of Einstein’s life was, therefore, quadratic in its error amplitude: one factor for introducing Λ for the wrong reason, the second factor for retracting Λ for the wrong reason! Henceforth, except for cosmological problems, we shall always set $\Lambda = 0$.

Exercise 5.9.1

Show that equation (5.67) may be written in source-function form as

$$R_{\mu\nu} = -\frac{8\pi G S_{\mu\nu}}{c^4} - \Lambda g_{\mu\nu}.$$

Because the field equations have been constructed to admit a new mathematical solution by a transformation of coordinates, they will generally admit more than one solution, unless additional constraints are provided. To make this point as clear as possible, imagine that we have worked hard and solved for the metric $g_{\mu\nu}$, and it turned out to be plain old Minkowski space.⁷ We had the usual time t -coordinate for index 0, and let us say (α, β, γ) for the three spatial dimensions. Even if we restricted ourselves to diagonal $g_{\mu\nu}$ only, we might have found that the four diagonal entries were $(-1, 1, 1, 1)$ or $(-1, 1, \alpha^2, 1)$ or $(-1, 1, \alpha^2, \alpha^2 \sin^2 \beta)$ depending upon whether we happened to be using Cartesian (x, y, z) , cylindrical (R, ϕ, z) , or spherical (r, θ, ϕ) spatial coordinate systems. The upside is that we always have the freedom to work with coordinates that simplify our equations, or make physical properties of our solutions more transparent, or join smoothly onto a favored set of flat-space coordinates at large distances from a source. The downside is that in a complicated problem, it is far from easy to know the best coordinates to use. For example, when numerically calculating the metric tensor for a dynamically complex system like merging black holes, one must solve for the coordinates as part of the evolving problem itself!

5.10 A Lorentz-invariant gravitational pseudotensor

Let us write the general metric tensor $g_{\mu\nu}$ in the form

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad (5.68)$$

where $\eta_{\mu\nu}$ is the Minkowski metric. This is a useful and obvious decomposition when $h_{\mu\nu}$ is small compared with unity, as would be the case for either the Newtonian or weak-radiation limits, but here we shall not restrict ourselves to this assumption. We shall assume only that $h_{\mu\nu} \rightarrow 0$ at large distances from the sources under consideration. Nevertheless, the linear (in $h_{\mu\nu}$) components of $G_{\mu\nu}$ are of interest in their own right. The theory of weak gravitational radiation is developed in chapter (9); here we use a similar formalism to construct expressions for the total mass/energy and angular momentum associated with the bound source of a static geometry. In the following section, we use this same approach to present an alternative, and very simple, route to the field equations.

The metric tensor $g_{\mu\nu}$ behaves as a tensor under any proper coordinate transformation, of course. While $h_{\mu\nu}$ is not itself a true tensor, the fact that $\eta_{\mu\nu}$ transforms identically under Lorentz transformations means that $h_{\mu\nu}$ must be at least Lorentz covariant, which implies

⁷If we are using ill-suited coordinates, it can be *very* hard to tell. We'll see an explicit example of this when we study cosmology.

that the partial derivatives of $h_{\mu\nu}$ are Lorentz-covariant tensors, as well. They are, in this sense, *pseudotensors*. The indices of $h_{\mu\nu}$ and its derivatives are raised and lowered by $\eta_{\mu\nu}$, preserving Lorentz covariance in all their forms.

We now expand the left side of the field equation in powers of $h_{\mu\nu}$, retaining the linear terms while placing quadratic and higher-order terms on the right side of the equation. Formally, we then have

$$R_{\mu\nu}^{(1)} - \frac{1}{2}\eta_{\mu\nu}R^{(1)} = -\frac{8\pi GT_{\mu\nu}}{c^4} - \sum_{n=2}^{\infty} \left(R_{\mu\nu}^{(n)} - \frac{\eta_{\mu\nu}}{2}R^{(n)} - \frac{h_{\mu\nu}}{2}R^{(n-1)} \right), \quad (5.69)$$

where the superscript (n) notation denotes n th powers of $h_{\mu\nu}$ grouped together. The field equations then take the form

$$R_{\mu\nu}^{(1)} - \frac{1}{2}\eta_{\mu\nu}R^{(1)} = -\frac{8\pi G\tau_{\mu\nu}}{c^4}, \quad (5.70)$$

with

$$\tau_{\mu\nu} = T_{\mu\nu} + \frac{c^4}{8\pi G} \sum_{n=2}^{\infty} \left(R_{\mu\nu}^{(n)} - \frac{\eta_{\mu\nu}}{2}R^{(n)} - \frac{h_{\mu\nu}}{2}R^{(n-1)} \right) \equiv T_{\mu\nu} + t_{\mu\nu}. \quad (5.71)$$

Equations (5.70) and (5.71) may at first appear to be just a formal reworking of the field equations, with all the nonlinear complexity subsumed in the artificial construct $t_{\mu\nu}$. But $t_{\mu\nu}$ has physical significance. First, $\tau_{\mu\nu}$ is Lorentz covariant, since it is proportional to $R_{\mu\nu}^{(1)}$ and its trace. Thus $t_{\mu\nu}$, the difference between two Lorentz-covariant quantities, must itself be Lorentz covariant. Second, $t_{\mu\nu}$ appears as a simple additive partner to $T_{\mu\nu}$ and captures all nonlinear terms of the field equations that are quadratic and higher order in the derivatives of the metric tensor. It is plausible, therefore, to identify $t_{\mu\nu}$ as the gravitational stress-energy pseudotensor (it respects Lorentz transformations but not general coordinate transformations) generated by the gravitational field *itself*, and we shall later strengthen this line of argument. This means, in turn, that the pseudotensor $\tau_{\mu\nu}$ represents the total stress-energy from both material and gravitational sources. Note that indices on both $t_{\mu\nu}$ and $\tau_{\mu\nu}$ are governed by $\eta_{\mu\nu}$, whereas only $g_{\mu\nu}$ can perform this role for the true tensor $T_{\mu\nu}$.

Following the linearized Bianchi identities, the ordinary divergence of $\tau_{\mu\nu}$ vanishes exactly:

$$\partial^\nu \tau_{\mu\nu} = \partial^\nu (R_{\mu\nu}^{(1)} - \eta_{\mu\nu}R^{(1)}/2) = 0. \quad (5.72)$$

Since the divergence of $\tau^{\mu\nu}$ vanishes within any spatial volume V , we have, in particular,

$$\frac{d}{dt} \int_V \tau^{0\nu} d^3x = - \int_V \partial_i \tau^{i\nu} d^3x = - \int_S \tau^{i\nu} n_i dS \rightarrow - \int_S t^{i\nu} n_i dS, \quad (5.73)$$

where we have used the standard form of the divergence theorem in the second equality, converting to a surface integral. The final term on the right relies on the assumption that $T_{\mu\nu}$ vanishes at finite radius (or at least falls off more rapidly than $1/r^2$), whereas a contribution

from $t^{\mu\nu}$ may be present in the form of time-dependent gravitational radiation. For static geometries however, $\tau^{\mu\nu}$ in the surface integral on the right will generally fall off sufficiently rapidly (typically $(\partial_\mu h^{\nu\lambda})^2 \sim 1/r^4$) that the $\tau^{0\nu}$ volume integral is time independent and convergent. When this condition is met, then by analogy with the ordinary stress tensor, the integral

$$P^\nu = \int_V \tau^{0\nu} d^3x \quad (5.74)$$

may be interpreted as the total energy-momentum pseudovector of matter, radiation, *and gravitation*. Despite this remarkable inclusiveness (more accurately, *because* of it), the integral in equation (5.74) is taken over ordinary Euclidean space.

One may also define the antisymmetric pseudotensor

$$J^{\mu\nu} = \int_V (x^\mu \tau^{0\nu} - x^\nu \tau^{0\mu}) d^3x, \quad (5.75)$$

whose spatial J^{ij} components now correspond to the combined angular momentum of gravitation, radiation, and matter. Further discussion and applications of P^ν and $J^{\mu\nu}$ to black hole geometries may be found in §11.2.

5.11 Field equations from local inertial coordinates

Let us now turn our problem around by directly evaluating the divergence of the Ricci tensor in local inertial coordinates. This is a very simple task but one that leads to very important consequences. We shall be able to construct a one-line proof of the contracted Bianchi identities and thereby an alternative derivation of the field equations themselves. The key point here is that in local inertial coordinates erected about a particular point X^ρ in spacetime, $h_{\mu\nu}$ is quadratically small in the coordinate difference $\delta x \sim x^\rho - X^\rho$. This means that *all nonlinear terms in $R_{\mu\nu} - g_{\mu\nu}R/2$ vanish in local inertial coordinates*—which are precisely the coordinates constructed to ensure that the influence of all local gravitational fields vanishes! This is a very compelling reason to identify these nonlinear geometric terms with the stress tensor of the gravitational field itself, as we did more informally in the previous section. Moreover, because the corrections to the leading-order Ricci terms are quadratic in δx , we have not only the local equality

$$R_{\mu\nu}(\text{local inertial coordinates}) = R_{\mu\nu}^{(1)} \quad (5.76)$$

but also local equality even *after taking the divergence*. More explicitly, if we define $h^\mu_\nu \equiv \eta^{\mu\rho} h_{\rho\nu}$, $h \equiv h^\mu_\mu$, then to leading order in $h_{\mu\nu}$ we have

$$\det g_{\mu\nu} = -(1 + h), \quad \Gamma^\lambda_{\mu\nu} = \frac{1}{2}(\partial_\mu h^\lambda_\nu + \partial_\nu h^\lambda_\mu - \partial^\lambda h_{\mu\nu}). \quad (5.77)$$

Note that the lowest-order neglected terms are of order $h^2 \sim (\delta x)^4$ in the left equation above, and $h \partial h \sim (\delta x)^3$ in the right equation. They may be safely ignored, even after the differentiations to follow.

Using the form of the Ricci tensor from page (xiv) appropriate to local inertial coordinates, we readily find

$$2R_{\mu\nu} = \partial_\mu \partial_\nu h - 2 \partial_\lambda \Gamma_{\mu\nu}^\lambda = \partial_\mu \partial_\nu h - \partial_\mu \partial_\lambda h^\lambda_\nu - \partial_\nu \partial_\lambda h^\lambda_\mu + \partial_\lambda \partial^\lambda h_{\mu\nu} = 2R_{\mu\nu}^{(1)}, \quad (5.78)$$

and

$$R = R^\mu_\mu = \partial_\mu \partial^\mu h - \partial_\mu \partial_\nu h^{\mu\nu} = R^{(1)}. \quad (5.79)$$

Raising the indices on $R_{\mu\nu}$ and taking the divergence, we find the second and fourth triple derivatives cancel in equation (5.78), leaving

$$2 \partial_\mu R^{\mu\nu} = \partial^\nu (\partial_\mu \partial^\mu h - \partial_\mu \partial_\lambda h^{\lambda\mu}) = \eta^{\mu\nu} \partial_\mu (\partial_\lambda \partial^\lambda h - \partial_\lambda \partial_\rho h^{\lambda\rho}) = \eta^{\mu\nu} \partial_\mu R. \quad (5.80)$$

This result leads immediately to the contracted Bianchi identities, $\partial_\mu (R^{\mu\nu} - \eta^{\mu\nu} R/2) = 0$, expressed in local inertial coordinates. Transforming the vanishing difference tensor to general coordinates, we obtain

$$\left(R^{\mu\nu} - \frac{g^{\mu\nu} R}{2} \right)_{;\mu} = 0.$$

With the fully covariant contracted Bianchi identities in hand, we know, by taking the Newtonian limit, that it is but a very short step to the general field equations. Indeed, a simple inspection of equation (5.78) immediately reveals the dominant $(\partial_i \partial^i h_{00})/2$ Laplacian term in R_{00} , which comprises the sole contribution to the left-side Ricci tensor of the static, weak-field equations in source-function form.

Finally, we note that by isolating the linear terms of the field equations in the formulation of equation (5.70), we have a better sense of a physical interpretation for the vanishing of the *covariant* divergence of $T_{\mu\nu}$ in the presence of spacetime curvature. This is in effect an interchange between the material stress-energy and gravitational stress-energy components. By contrast, the vanishing of the ordinary divergence of $\tau_{\mu\nu}$ expresses a true local conservation of the combined material *plus* gravitational stress-energy. There is nowhere else to go.

5.12 Variational derivation of the field equations*

It is a remarkable and undisputed fact of scientific history that priority for the first derivation of the gravitational field equations belongs not to Einstein but to the mathematician David Hilbert, who presented it in a talk five days in advance of Einstein's well-known lecture of November 25, 1915. While Hilbert freely acknowledged that the physical content of relativistic gravity theory was entirely due to Einstein, as a leading mathematician, Hilbert was in a better position to bring variational methods to bear on this problem. At the time, it was already understood that such techniques are generally the quickest and most powerful route for deriving fundamental field equations in physics, affording foundational insights which are otherwise obscured in more detailed derivations. In this section, we present a brief introduction to the variational approach to the gravitational field equations. These techniques are widely used today as a means for investigating possible modifications of

general relativity while remaining consistent with the fundamental principles of relativity theory.

5.12.1 Matter stress tensor from a variational principle

In §(3.4) we showed that the geodesic equations of motion may be found by maximizing

$$\int \sqrt{-g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu} d\tau$$

between two fixed points in spacetime. (The overdot notation denotes $d/d\tau$, as usual.) Motivated by this finding, let us define the quantity I_M , the matter action, for an ensemble of particles with rest masses m_n and coordinates $x_n^\mu(\tau)$ by

$$I_M = - \sum_n m_n c^2 \int \sqrt{-g_{\mu\nu} \dot{x}_n^\mu \dot{x}_n^\nu} d\tau. \quad (5.81)$$

The lead minus sign is inserted for later convenience. By varying the coordinates $x_n \rightarrow x_n + \delta x_n$ and demanding $\delta I_M = 0$, we have seen that the individual particle geodesic equations emerge. We shall now show that if instead of varying x_n , we vary $g_{\mu\nu}$, and write the resulting variation in I_M as

$$\delta I_M = \frac{1}{2} \int T^{\mu\nu} \delta g_{\mu\nu} \sqrt{g} d^4x, \quad (5.82)$$

then the quantity $T^{\mu\nu}$ defined by this expression is, in fact, the true stress-energy tensor for the ensemble of mass particles. Equation (5.82) is, in fact, a very general method for finding the stress-energy tensor for both particles *and* fields.

In equation (2.61), we wrote an expression for the particle stress tensor by averaging over the direct products of their individual 4-velocities and multiplying by a rest density. We require here a more fine-scale expression for the stress-energy tensor, and we shall not impose the requirement that the ensemble of particles be at rest on average. In special relativity, this fine-scale matter stress tensor takes the form

$$T_M^{\alpha\beta} = \sum_n m_n \frac{\delta^3(x - x_n)}{\gamma_n} \dot{x}_n^\alpha \dot{x}_n^\beta. \quad (5.83)$$

The combination of $m_n \delta^3(x - x_n)$ in an expression for a fine-scale mass density is clear enough, but the factor of γ_n may at first seem strange. This stems from the fact that $\delta^3(x - x_n)$ is by itself not a Lorentz invariant, for much the same reason the volume d^3x is not a Lorentz invariant. By contrast, $dt d^3x = \gamma d\tau d^3x$ is a Lorentz invariant, which means that γd^3x is Lorentz invariant. Since the pure number $\delta^3(x - x_n) d^3x$ is manifestly a Lorentz invariant, we conclude that $\delta^3(x - x_n)/\gamma_n$ must also be Lorentz invariant. In effect, the γ_n denominator reduces the delta function to the individual particle rest-frame value, just as we were careful to do in defining ρ_r in equation (2.61).

Since $\gamma_n d\tau = dt$, equation (5.83) may be written as

$$T_M^{\alpha\beta} = \sum_n m_n \int \delta^4(x - x_n) \frac{dx_n^\alpha}{d\tau} \frac{dx_n^\beta}{dt} c dt, \quad (5.84)$$

by the addition of a delta function integration over $dx^0 = c dt$. In a manifestly *generally* covariant form, this expression becomes

$$T_M^{\mu\nu} = \frac{c}{\sqrt{g}} \sum_n m_n \int \delta^4(x - x_n) \frac{dx_n^\mu}{d\tau} \frac{dx_n^\nu}{d\tau} d\tau. \quad (5.85)$$

The leading $g^{-1/2}$ arises because the δ^4 function needs to acquire this factor when upgrading from Minkowski to general coordinates. (It may be taken out of the integral because it depends only upon coordinates x , whereas the integral is over τ in the $x_n(\tau)$ particle positions.) If we now vary equation (5.81) with respect to $g_{\mu\nu}$, we obtain

$$\begin{aligned} \delta I_M &= \frac{1}{2} \sum_n m_n c^2 \int (-g_{\mu\nu} \dot{x}_n^\mu \dot{x}_n^\nu)^{-1/2} \dot{x}_n^\mu \dot{x}_n^\nu \delta g_{\mu\nu} d\tau \\ &= \frac{1}{2} \sum_n m_n c \int \dot{x}_n^\mu \dot{x}_n^\nu \delta g_{\mu\nu} d\tau \\ &= \frac{1}{2} \int T_M^{\mu\nu} \delta g_{\mu\nu} \sqrt{g} d^4x, \end{aligned} \quad (5.86)$$

in agreement with equation (5.82).

5.12.2 The gravitational action

The Hilbert action, from which the gravitational/geometric contribution to the Einstein field equations may be derived, is both simple and elegant:

$$I_G = -\frac{c^4}{16\pi G} \int R \sqrt{g} d^4x, \quad (5.87)$$

where R is the Ricci scalar. In the absence of matter, the integral of this curvature scalar is a minimum when the Einstein vacuum field equations are satisfied, a sort of minimization of the externally forced distortion of Minkowski spacetime. To show this, we first recall equation (4.68) in the form appropriate to a variation in $\sqrt{g}$:

$$\delta \sqrt{g} = \frac{\sqrt{g}}{2} g^{\mu\nu} \delta g_{\nu\mu}, \quad (5.88)$$

and second, the Palatini identity in the form of equation (5.42):

$$\sqrt{g} g^{\mu\nu} \delta R_{\mu\nu} = \frac{\partial}{\partial x^\nu} \left[\sqrt{g} \left(g^{\mu\nu} \delta \Gamma_{\mu\lambda}^\lambda - g^{\mu\lambda} \delta \Gamma_{\mu\lambda}^\nu \right) \right]. \quad (5.89)$$

Then, varying the integrand of (5.87), we obtain

$$\begin{aligned}
 \delta(\sqrt{g}R) &= \frac{\sqrt{g}}{2} g^{\mu\nu} \delta g_{\nu\mu} R + \sqrt{g} \delta(g^{\mu\nu} R_{\mu\nu}), \\
 &= \frac{\sqrt{g}}{2} g^{\mu\nu} \delta g_{\nu\mu} R + \sqrt{g} R_{\mu\nu} \delta g^{\mu\nu} + \underbrace{\sqrt{g} g^{\mu\nu} \delta R_{\mu\nu}}_{\text{zero contribution}}, \\
 &= \frac{\sqrt{g}}{2} g^{\mu\nu} \delta g_{\nu\mu} R - \sqrt{g} R_{\mu\nu} g^{\mu\kappa} g^{\nu\lambda} \delta g_{\kappa\lambda}.
 \end{aligned} \tag{5.90}$$

The last term in the second line of the above makes no contribution, because it is a pure divergence term, which vanishes at infinity when the integral is performed. The third line follows directly from $\delta(g^{\mu\nu} g_{\nu\kappa}) = 0$. Then, carrying through the index raising on $R_{\mu\nu}$ in the second term leads to

$$\delta I_G = \frac{c^4}{16\pi G} \int \left(R^{\mu\nu} - \frac{g^{\mu\nu}}{2} R \right) \delta g_{\mu\nu} \sqrt{g} d^4x. \tag{5.91}$$

Finally, setting the variation of the total action $\delta I_M + \delta I_G = 0$, we find

$$R^{\mu\nu} - \frac{g^{\mu\nu}}{2} R + \frac{8\pi G}{c^4} T^{\mu\nu} = 0, \tag{5.92}$$

the Einstein field equations: for Professor Hilbert, a straightforward, post-lecture exercise.

Exercise 5.12.1

Is $\delta g_{\mu\nu}$ a true tensor? Explain.

Exercise 5.12.2

Show that a cosmological constant Λ can be accommodated in the variational formalism by replacing R with $R + 2\Lambda$ in the integral I_G .

Exercise 5.12.3

Show that by varying $g_{\mu\nu} \rightarrow g_{\mu\nu} + \delta g_{\mu\nu}$, the action integral

$$\int \left[\left(\frac{3P}{2} + \frac{\rho c^2}{2} \right) + \frac{1}{2} \left(\rho + \frac{P}{c^2} \right) U^\mu U^\nu g_{\mu\nu} \right] \sqrt{g} d^4x$$

yields the stress tensor for an ideal gas,

$$T^{\mu\nu} = Pg^{\mu\nu} + (\rho + P/c^2)U^\mu U^\nu,$$

when equation (5.82) is applied.

5.12.3 Conservation of $T^{\mu\nu}$

An important feature of equation (5.82) is that the stress tensor thereby defined must be conserved: its covariant divergence must vanish. To see this, we reason (rather delicately) as follows.

The integral I_M is a scalar and is therefore unchanged by the formal coordinate transformation $x^\mu \rightarrow x'^\mu$. The various vector and tensor components that $T^{\mu\nu}$ comprises will change by the standard rules for coordinate transformations, as will $g_{\mu\nu}$, but the integral I_M will remain unchanged. Remember, however, that x'^μ is also just an ordinary integration variable. In carrying through the x' integration, we are allowed to use whatever variables we like. We could use standard calculus to change the variables from x'^μ back to x^μ , which would not change I_M —even if we were ignorant of the fact that our integrand actually involves formal vectors and tensors! To be as clear as possible on this point, let us suppose we used the formal vector transformation law to change the vector dynamic variable in I_M from $V^\mu(x^\nu)$ to $V'^\mu(x'^\nu)$, with $x'^\nu = f^\nu(x^\mu)$. Now, replacing the $V'^\mu(x'^\nu)$ components with $V'^\mu(f^\nu[x])$ by way of a standard change of variables, will not alter the value of I^M from its original x_μ system value. The first change of coordinates causes no change, because I_M is a coordinate scalar under formal transformation laws; the second causes no change because it is just an ordinary calculus change of variables. Note that this is true even though the difference $V'^\mu(f^\nu[x]) - V^\mu(x)$ will not, in general, vanish. This has consequences. Recall that in the course of deriving the equations of motion by a variational principle, we required that the net first-order changes in I_M from all dynamic variable changes must *vanish*. (Be careful not to confuse x_n^μ , which is the position of particle n and thus a dynamic variable, with x^μ , a mere coordinate.) Thus, by direct construction no surviving terms in δI_M can be found among the first-order cumulative changes in these dynamic n -variables. The sole constraint for $\delta I_M = 0$ that needs to be imposed arises exclusively from the first-order changes in $\delta g_{\mu\nu}$ created by our coordinate meddling: such changes must vanish on their own. And that means we can yet squeeze something more of interest out of the stationary nature of I_M .

In particular, let us consider the class of metric variations $\delta g_{\mu\nu}$ created by an infinitesimal coordinate transformation, $x'^\mu = f(x^\mu) = x^\mu + \epsilon^\mu(x)$, where the ϵ^μ are arbitrary but vanishingly small functions. Then,

$$g'_{\mu\nu}(x') = \frac{\partial x^\kappa}{\partial x'^\mu} \frac{\partial x^\lambda}{\partial x'^\nu} g_{\kappa\lambda}(x). \quad (5.93)$$

By definition, $\delta g_{\mu\nu} = g'_{\mu\nu}(x) - g_{\mu\nu}(x)$: the transformed metric tensor evaluated at the original x -coordinate point, minus the original metric at the very same point.⁸ With

⁸Knowledgeable readers will recognize here the Lie differential.

$$g_{\mu\nu}(x) \equiv \frac{\partial x^\kappa}{\partial x'^\mu} \frac{\partial x^\lambda}{\partial x'^\nu} g_{\kappa\lambda}(x) + g_{\mu\nu}(x) - g'_{\mu\nu}(x'), \quad (5.94)$$

we substitute $x^\mu = x'^\mu - \epsilon^\mu$ in the partial derivatives of the transform and expand this equation to leading linear order in ϵ^μ . We then find:

$$\delta g_{\mu\nu} = -g_{\kappa\nu} \partial_\mu \epsilon^\kappa - g_{\mu\kappa} \partial_\nu \epsilon^\kappa - \epsilon^\kappa \partial_\kappa g_{\mu\nu}. \quad (5.95)$$

Continuing from equation (5.82), we obtain

$$\begin{aligned} \delta I_M &= -\frac{1}{2} \int T^{\mu\nu} \sqrt{g} (g_{\kappa\nu} \partial_\mu \epsilon^\kappa + g_{\mu\kappa} \partial_\nu \epsilon^\kappa + \epsilon^\kappa \partial_\kappa g_{\mu\nu}) d^4x \\ &= \int [\partial_\mu (T^\mu_\kappa \sqrt{g}) - (\sqrt{g}/2) T^{\mu\nu} \partial_\kappa g_{\mu\nu}] \epsilon^\kappa d^4x \\ &= \int T^\mu_{\kappa;\mu} \epsilon^\kappa \sqrt{g} d^4x. \end{aligned} \quad (5.96)$$

The second line follows from the first by integration by parts of the first two terms, and the final line makes use of equation (4.74) (and exercise 4.3.2 that follows). We conclude that the stress-energy tensor $T^{\mu\nu}$ defined in this way must quite generally be conserved. If δI_M vanishes and ϵ^κ is arbitrary, then $T^\mu_{\kappa;\mu} = 0$.

The variational technique for deriving the field equations is powerful and opens the door to possible modifications of Einstein's relativistic theory of gravity which are fully covariant. The curvature scalar R in I_M could, in principle, be extended by a more general Taylor series expansion in R : $R + \alpha R^2 + \dots$, or by the introduction of yet more exotic curvature scalars, like the Kretschmann scalar $R^{\mu\nu\kappa\lambda} R_{\mu\nu\kappa\lambda}$. Unlike the inclusion of a cosmological constant Λ , which alters the field equations at large spacetime scales, higher-order terms in R alter the theory at small spacetime scales. Gravity is far from well explored at these small scales; hence, the possible relevance of these terms in an extended theory of gravity is very much a topic of current theoretical interest.